Question 2.1

i) The risk-neutral probability, p, is calculated as:

p = (1 + r - d) / (u - d)=(1.05 - 0.90) / (1.15 - 0.90) = 0.15 / 0.25 = 0.60

Construct the Binomial Tree for Stock Prices Over 2 Years

The resulting stock price tree is:

Year	Path	Stock Price
0	Initial	100
1	Up	115
1	Down	90
2	Up-Up	132.25
2	Up-Down	103.5
2	Down-Up	103.5
2	Down-Down	81

The payoff for a European put option at maturity is max(K - S, 0). So, the payoffs at maturity are

Path	Stock Price	Put Payoff
Up-Up	132.25	0
Up-Down	103.5	0
Down-Up	103.5	0
Down-Down	81	19

Thus the value of the put options is ((1 - p) (1 - p) x 19)/ (1+rf)^2=0.4 0.4 19/1.1025=2.757

i) One possibility is to go long in a Straddle. A straddle is the combination of a long position in a put and a call with the same strike price and expiration date. The profit of a long position in a straddle is positive when the stock price is far away from the strike price. If you expect the stock to end up near the strike price you could choose to sell a straddle. Another strategy could be to go long in a butterfly spread. This combination of two long call and two short call options pays off when the stock price is close to the strike price.

Question 2.2

i) NPV

NPV = 300 - 250 = 50 million NOK

Decision Based on the NPV Rule:
The project has a positive NPV of 50 million NOK, indicating that it is profitable. Based on the NPV rule, you should proceed with the project.

ii) Value of the Project as an Option Using the Black-Scholes Model

The project is interpreted as a European-style call option. The parameters are:

- $S = 300$ million NOK (value of cash inflows).
- $K = 250$ million NOK (exercise price).
- $T = 3$ years (time to expiration).
- $\sigma = 15\%$ or 0.15 (volatility of cash inflows).
- $r = 5\%$ or 0.05 (risk-free rate).

To calculate the option value:

1. The discounted exercise price is $K \times 1 / (1 + r)^T = 250 \times 0.8638 = 215.95$ million NOK.
2. The log term is $\ln(S / K_{discounted}) = \ln(300 / 215.95) = \ln(1.388) \approx 0.3286$.
3. The volatility term is $\sigma \times \sqrt{T} = 0.15 \times \sqrt{3} \approx 0.2598$.
4. $d1 = (\ln(S / K_{discounted}) + \sigma^2 \times T / 2) / (\sigma \times \sqrt{T}) = (0.3286 + 0.03375) / 0.2598 \approx 1.393$.
5. $d2 = d1 - \sigma \times \sqrt{T} = 1.393 - 0.2598 \approx 1.133$.

Using cumulative normal distribution probabilities:

- $N(d1) \approx 0.918$
- $N(d2) \approx 0.872$

The option value is $C = S \times N(d1) - K_{discounted} \times N(d2) = 300 \times 0.918 - 215.95 \times 0.872 \approx 87.26$ million NOK.

---

iii) Comparison of Results and Explanation

The value of the project as an option (87.26 million NOK) is higher than its NPV (50 million NOK).

Reason for the Difference:
The Black-Scholes option pricing model accounts for the **flexibility to wait** before committing to the project. In this case, the 3-year government permit gives the company the ability to delay the investment until market conditions are more favorable. The NPV method does not capture this flexibility, while the option value reflects the additional value of this strategic choice.

Question 3.1

EcoNor's current stock price:

- Total market value of assets = $600 million
- Total debt = $250 million
- Total equity value = $600 million - $250 million = $350 million
- Shares outstanding = 7 million

Current stock price = Total equity value / Shares outstanding
= $350 million / 7 million = $50 per share

EcoNor's current stock price is $50 per share.

Stock price after the share repurchase:

- Cash distributed = $70 million
- New total market value of assets = $600 million - $70 million = $530 million
- New equity value after repurchase = $530 million - $250 million = $280 million
- Shares remaining (before calculating reduction) = 7 million

New stock price = New equity value / Shares remaining
= $280 million / 7 million = $50 per share

EcoNor's stock price after the repurchase will remain $50 per share.

Number of shares outstanding after the repurchase:

- Cash used for repurchase = $70 million
- Repurchase price per share = $50
- Number of shares repurchased = $70 million / $50 = 1.4 million shares

Shares outstanding after repurchase = 7 million - 1.4 million = 5.6 million shares

The number of shares outstanding after the repurchase will be 5.6 million.

Quesiton 3.2

- **On the ex-dividend date, what should happen to EcoNor's share price, and why?**
  - Since EcoNor is distributing $70 million as a dividend, the company's market value will decrease by the amount of the dividend payout. With 7 million shares outstanding, the dividend per share is calculated as follows:

    Dividend per share = Total dividend / Shares outstanding
    = $70 million / 7 million = $10 per share

  - On the ex-dividend date, EcoNor's share price should decrease by $10 per share to reflect this payout, as shareholders lose claim to the $10 per share dividend once the stock trades ex-dividend.

  Therefore, EcoNor's share price should drop by $10 per share on the ex-dividend date.

- **How an investor could earn a profit in two scenarios:**
  - **If the stock price changes by less than the amount found in (i):**

    If EcoNor's stock price decreases by less than $10 per share on the ex-dividend date, an investor could earn a profit by buying the stock just before

it goes ex-dividend and selling it just after, as the dividend would more than cover the capital loss on the stock.

- **If the stock price changes by more than the amount found in (i):**

  If EcoNor's stock price decreases by more than $10 per share, the stock would be priced lower than expected. An investor could earn a profit by selling the stock just before it goes ex-dividend and buying it just after.

Question 3.3

The investor would be indifferent between dividends and share repurchases if they prioritize cash, as either method allows them to achieve the same total wealth.

Scenario 1: EcoNor Distributes $70 Million as a Dividend

Dividend per Share: $70 million / 7 million shares = $10 per share.

Investor's Cash from Dividend: 100 shares x $10 = $1,000.

Stock Price on Ex-Dividend Date: $50 - $10 = $40.

Total Wealth: The investor holds $1,000 in cash and 100 shares at $40 each, totaling $5,000.

Scenario 2: EcoNor Uses the $70 Million for Share Repurchase

Repurchase Price: Shares are repurchased at $50 each.

Investor's Cash by Selling Shares: The investor can sell 20 shares at $50, generating $1,000 in cash.

Remaining Shares: The investor keeps 80 shares at $50 each.

Total Wealth: The investor holds $1,000 in cash and 80 shares valued at $50 each, also totaling $5,000.

Quesiton 4.1

i)

Using Decker's P/E, we would estimate a share price for RCKY of Po = $2.30 × 13.3 = $30.59.

Using the enterprise value to EBITDA multiple, we would estimate RCKY's enterprise value to be Vo = $30.7 million x 7.4 = $227.2 million.

We then subtract debt and divide by the number of shares to estimate RCKY's share price: Po = (227.2 - 125)/5.4 = $18.93.

ii) Because of the large difference in leverage between the firms, we would expect the second estimate, which is based on enterprise value, to be more reliable.

iii) Some example here. Any 2 reasonable arguments would work.

Dependence on Peer Group Selection: The reliability of multiples depends on selecting truly comparable companies. If the peer group differs in terms of size, growth stage, or business mix, the resulting valuation might not accurately reflect the target company's fair value.

Lack of Company-Specific Detail: Multiples provide a broad, market-based valuation without accounting for unique factors of the specific company, such as its growth potential, risk profile, or operational efficiencies. This can lead to over- or undervaluation if the company differs significantly from its peers in these aspects.

Influence of Accounting Differences: Variations in accounting practices across companies, such as differing depreciation methods, inventory valuations, or revenue recognition policies, can distort multiples. This makes it difficult to compare companies accurately, as reported earnings or book values may not reflect true economic value consistently.

Question 4.2

i) Sensitivity Analysis on GKB's Estimated Share Price

Original P/E Multiple of 13.3 Estimated share price = $30.59

P/E Multiple Increased by 10% to 14.63 Estimated share price = $33.65

P/E Multiple Decreased by 10% to 11.97 Estimated share price = $27.53

ii) Discus the Sensitivity and Reliability of the P/E-Based Valuation

The sensitivity analysis shows that a 10% change in the P/E multiple leads to a range of GKB's estimated share price from $27.53 to $33.65. This represents a change of approximately ±10% around the original estimate of $30.59.

This sensitivity indicates that GKB's valuation is quite dependent on the assumed P/E multiple. A small change in the multiple has a relatively significant impact on the estimated share price, which suggests that the P/E-based valuation may be less reliable if the comparable P/E ratio is uncertain or varies widely. In such cases, other valuation methods or additional comparables may be necessary to cross-check the estimate.

Question 4.3

Disregard Questions 4.1 and 4.2. Consider now Xpertise AI, a company claiming to have mastered everything from social media to space exploration algorithms. It is expected to generate a free cash flow of $8 million in the coming year, with this cash flow projected to grow at 3% annually. Xpertise AI has an equity cost of capital of 13%, a debt cost of capital of

7%, and faces a 21% corporate tax rate. The company also maintains a constant debt-to-equity ratio of 0.5.

i) Calculate Xpertise AI's Weighted Average Cost of Capital (WACC).

ii) Determine the value of Xpertise AI as an unlevered company, ignoring the effects of debt.

iii) Calculate the value of Xpertise AI's interest tax shield.

Tasks:

Solution

i) E/V = 1 / (1 + 0.5) = 0.6667; D/V = 0.5 / (1 + 0.5) = 0.3333

WACC = (E/V) * Re + (D/V) * Rd * (1 - Tc)= (0.6667 * 0.13) + (0.3333 * 0.07 * (1 - 0.21))= 0.08667 + 0.02343 = 0.1051, or 10.51%

ii)

r_u = (E/V) * Re + (D/V) * Rd = (0.6667 * 0.13) + (0.3333 * 0.07) = 0.11, or 11%

Using the perpetuity growth model for an unlevered firm:

V_U = FCF / (R_u - g)

Substituting the values:

V_U = 8,000,000 / (0.11 - 0.03) = 8,000,000 / 0.08 = 100,000,000

(iii): Calculate the Value of Xpertise AI's Interest Tax Shield

The interest tax shield is the additional value created by debt due to tax savings on interest payments. We can calculate this in two ways: as the difference between the levered and unlevered firm values or by directly discounting the tax savings.

Method 1: Difference Between Levered and Unlevered Firm Values

1. **Calculate the Levered Firm Value (V_L) of Xpertise AI**

   V_L = FCF / (WACC - g)= 8,000,000 / (0.1051 - 0.03) = 8,000,000 / 0.0751 = 106,524,634

2. **Calculate the Interest Tax Shield**

   The interest tax shield is the difference between the levered and unlevered firm values:

   Interest Tax Shield = V_L - V_U = 106,524,634 - 100,000,000 = 6,524,634

Method 2: Direct Discounting of Tax Savings

Alternatively, we can calculate the interest tax shield as a growing perpetuity of the annual tax savings on debt interest, discounted by the unlevered cost of capital (R_u).

The formula for the interest tax shield is:

Interest Tax Shield = $(D * Rd * Tc) / (R_u - g)$

Since we know the debt-to-equity ratio is 0.5, and we have already calculated the levered firm value V_L, we can calculate the debt amount as follows:

D = Debt Ratio * V_L

Substituting the values:

D = 0.3333 * 106,524,634 = 35,508,211

1. **Substitute into the Tax Shield Formula**

   Now we can calculate the interest tax shield:

   Interest Tax Shield = (35,508,211 * 0.07 * 0.21) / (0.11 - 0.03) = 521,973.1 / 0.08 = 6,524,634